BMO BANK CREDIT RISK ANALYTICS

Credit Card & Personal Loan Portfolio - Complete Power BI Implementation Guide

Senior Analyst / Analytics Lead BAU Documentation
Synthetic Portfolio (Industry-Representative): $12.7B | 30,000 Accounts | 27+ Months
With 20 Real Interview Q&A

📋 PROJECT OVERVIEW

Client & Engagement

- Bank: Bank of Montreal (BMO) - Major Canadian Institution
- Consulting Partner: QUAUTO Technologies (D2PR Analytics Division)
- YOUR Role: Senior Data Analyst / Analytics Lead
- Duration: August 2023 - Present (27+ months, ongoing)
- Portfolio Scope: Credit Card (60%) & Personal Loan (40%) ONLY
  - Credit Card accounts: 18,000 | Personal Loan accounts: 12,000
  - Total portfolio (synthetic, representative): $12.7 Billion
  - Delinquent accounts: 4,200 (~14% delinquency rate, industry-realistic)
  - Annual charge-offs: 220–300 accounts (~1–2% charge-off rate)

IMPORTANT DISCLAIMER

All portfolio numbers are synthetic and generated to match industry norms, not actual BMO confidential data. The architecture, logic, and processes are based on your real Capgemini experience; the account/balance figures are representative for demonstration and interview preparation purposes.

👥 TEAM STRUCTURE & YOUR SPECIFIC WORK

YOUR ROLE (Indresh Tiwari - Senior Analyst / Analytics Lead)

Work Allocation:

- 40% Strategic Analytics & Architecture (20 hrs/week)
  - Design star schema data model (5 fact tables, 6 dimension tables)
  - Define delinquency bucket logic (Current → X+1 → X+5 → Charge-Off)
  - Build BCR formula and risk segmentation algorithms
  - Power BI data model, relationships, refresh logic

- 25% Stakeholder Management (12.5 hrs/week)
    - Daily (9–11 AM ET / 6:30–8:30 PM IST): Ad-hoc queries, dashboards access
    - Weekly: Portfolio reviews with SVP Management, CRO
    - Monthly: Executive presentations, board-level insights
- 20% Team Leadership & Mentorship (10 hrs/week)
    - Lead 5 junior/mid-level analysts (Rahul, Priya, Amit, Deepak, Neha)
    - Define task assignments, code reviews, quality gates
    - Rahul promoted to Mid-level in 6 months; Priya now leads Collections initiative
- 10% Data Governance & Compliance (5 hrs/week)
    - BCBS 239 risk data framework
    - Fair lending compliance (FICO risk-adjusted scoring)
    - Data quality audits (target 2% error rate, achieved)
- 5% Operational Delivery (2.5 hrs/week)
    - Dashboard refresh schedules, SLA monitoring (99.5% uptime)
    - 100+ daily users, 1000+ views per week
    - Incident resolution, user support

TEAM MEMBER #1: Rahul Sharma (Mid-level Analyst - Origination)

- CAD (Captured Application Details) data ingestion & validation
- Application volume trends, approval rates by product/underwriter
- Underwriter performance scoring
- Weekly Origination Executive Dashboard updates
- Tools: SQL, Power BI, Alteryx

TEAM MEMBER #2: Priya Desai (Junior Analyst → Collection Lead)

- Collection activity monitoring and workflows
- Collector performance scorecards (risk-adjusted)
- Priority scoring automation (Balance × Bucket Weight × FICO Factor)
- Daily collection reports to ICU/Agency teams
- Tools: Power BI, Alteryx, SQL
- ACHIEVEMENT: Promoted after demonstrating expertise; now leads new collection optimization initiative

TEAM MEMBER #3: Amit Kumar (Junior Analyst - Performance)

- Delinquency bucket calculations (supports YOUR design)
- Portfolio health KPIs (BCR, GLPA, BLPA, cure rates)

- Payment history pattern analysis
- Monthly cohort reports (vintage analysis)
- Tools: SQL, Power BI

TEAM MEMBER #4: Deepak Singh (Junior Analyst - Ad-hoc)

- Executive ad-hoc analysis requests
- Strategic deep-dives (e.g., Q1 seasonality analysis)
- Trend analysis, risk modeling support
- Tools: Power BI, SQL, Excel

TEAM MEMBER #5: Neha Gupta (Junior Analyst - SAS/Legacy Systems)

- SAS script maintenance (portfolio risk models)
- Legacy system monitoring (pre-Snowflake data)
- Data quality checks (null%, duplicates, format)
- Tools: SAS, SQL, Data validation scripts

SUPPORTING ROLES (Shared with team)

- SQL/BI Developer: Query optimization, Snowflake Bronze-Silver-Gold layer management, performance tuning
- Data Scientist: Portfolio risk modeling, predictive analytics (SAS environment), vintage analysis
- Reporting Specialist: Dashboard monitoring, user SLA support, refresh logs
- Data Engineer: Alteryx automation workflows, BigQuery sync, Snowflake infrastructure

🛠 TECHNOLOGY STACK & TOOL-SPECIFIC WORK

ALTERYX (Daily ETL/Data Blending - Fully Automated)

YOUR Role: Design workflows, define data quality checks, schedule automation

Daily Workflow (6:00 AM IST / 8:30 PM ET previous day, every Mon–Fri):

text

```
Step 1: Read & Validate (6:00-6:15 AM IST)
├─ Input: 12 Excel data marts
├─ Batch macro pulls from shared folder
```

```
├─ Data quality checks:
│  ├─ Completeness: null% < 0.1%
│  ├─ Duplicates: account_id uniqueness validated
│  ├─ Format: dates YYYY-MM-DD, amounts numeric, fico 300-850
│  └─ Error flags logged if violations detected
└─ Output: Raw data → Snowflake Bronze layer

Step 2: Data Standardization (6:15–6:30 AM IST)
├─ Null handling (replace blanks in optional fields)
├─ Value standardization (e.g., "Current" → "Current/X", "CCcard" →
"Credit Card")
├─ Duplicate removal (keep latest snapshot date)
└─ Record ID assignment (unique row identifier)

Step 3: Dimensional Joins (6:30–6:45 AM IST)
├─ Join: CAD → Customer, Bureau → Customer
├─ Join: Delinquent Accounts → Customer, Collector
├─ Join: Chargeoff → Recovery Activity
└─ Output: Conformed fact tables

Step 4: Quality Gate (6:45–7:00 AM IST)
├─ Row count checks (compare yesterday vs today)
├─ Summary stats (min/max/avg balance, app counts)
├─ Alert if variance > 2%
└─ Email alert to YOU with summary

Send to Snowflake Bronze layer at 7:00 AM IST
```

Weekly Workflows (Monday 12 PM IST / 11:30 PM ET Sunday):

- Delinquency bucket calculation (group accounts by DPD ranges)
- BCR aggregation (GLPA/BLPA by product/region)
- Collector assignment automation (priority-based distribution)

SQL / SNOWFLAKE (Data Transformation - Bronze → Silver → Gold)

YOUR Role: Define transformation logic, create aggregations, optimize queries

Daily SQL Scripts (8:00–10:00 AM IST / 11:30 PM–1:30 AM ET prev day):

Script 1: Delinquency Bucket Calculation

sql

```sql
CREATE TABLE silver.account_delinquency_current AS
SELECT
    account_id,
    customer_id,
    product_type,
    outstanding_balance,
    CASE
        WHEN days_past_due <= 30 THEN 'Current/X'
        WHEN days_past_due BETWEEN 31 AND 60 THEN 'X+1'
        WHEN days_past_due BETWEEN 61 AND 90 THEN 'X+2'
        WHEN days_past_due BETWEEN 91 AND 120 THEN 'X+3'
        WHEN days_past_due BETWEEN 121 AND 150 THEN 'X+4'
        WHEN days_past_due BETWEEN 151 AND 180 THEN 'X+5'
        ELSE 'Charge-Off'
    END AS current_bucket,
    LAG(current_bucket) OVER (PARTITION BY account_id ORDER BY
snapshot_date) AS previous_bucket,
    CURRENT_DATE AS snapshot_date
FROM bronze.account_performance
WHERE snapshot_date = CURRENT_DATE;
```

Script 2: BCR Calculation (Balance Control Ratio)

sql

```sql
CREATE TABLE silver.portfolio_bcr AS
SELECT
    DATE_TRUNC('month', snapshot_date) AS report_month,
    product_type,
    -- GLPA: Good Loan Per Account
    SUM(CASE WHEN current_bucket = 'Current/X' THEN
outstanding_balance ELSE 0 END)
    / NULLIF(COUNT(DISTINCT CASE WHEN current_bucket = 'Current/X'
THEN account_id END), 0) AS glpa,

    -- BLPA: Bad Loan Per Account
    SUM(CASE WHEN current_bucket IN
('X+1','X+2','X+3','X+4','X+5','Charge-Off') THEN outstanding_balance
```

```
ELSE 0 END)
    / NULLIF(COUNT(DISTINCT CASE WHEN current_bucket IN
('X+1','X+2','X+3','X+4','X+5','Charge-Off') THEN account_id END), 0)
AS blpa,

    -- BCR Ratio (Target: 3.5-4.5)
    ROUND(
        SUM(CASE WHEN current_bucket = 'Current/X' THEN
outstanding_balance ELSE 0 END) /
        SUM(CASE WHEN current_bucket IN
('X+1','X+2','X+3','X+4','X+5','Charge-Off') THEN outstanding_balance
ELSE 0 END),
        2
    ) AS bcr_ratio,

    COUNT(DISTINCT account_id) AS total_accounts,
    SUM(outstanding_balance) AS total_balance
FROM silver.account_delinquency_current
GROUP BY DATE_TRUNC('month', snapshot_date), product_type
ORDER BY report_month DESC, product_type;
```

Script 3: Collection Priority Score

```sql
CREATE TABLE silver.collection_priority_score AS
SELECT
    account_id,
    customer_id,
    outstanding_balance,
    delinquency_bucket,
    fico_score_current,
    -- Priority = (Balance × Bucket Weight × FICO Factor)
    LEAST(100, CEIL(
        (outstanding_balance / 10000.0) *
        CASE
            WHEN delinquency_bucket = 'X+1' THEN 1
            WHEN delinquency_bucket = 'X+2' THEN 2
            WHEN delinquency_bucket = 'X+3' THEN 3
            WHEN delinquency_bucket = 'X+4' THEN 4
```

```
            WHEN delinquency_bucket = 'X+5' THEN 5
            ELSE 1
        END *
        CASE
            WHEN fico_score_current < 550 THEN 1.5
            WHEN fico_score_current BETWEEN 550 AND 700 THEN 1.2
            ELSE 1.0
        END
    )) AS priority_score
FROM silver.account_delinquency_current
ORDER BY priority_score DESC;
```

SAS (Legacy Portfolio Analytics - Maintained by Senior Data Scientist)

YOUR Role: Monitor outputs, incorporate into dashboards, alert on anomalies

Weekly SAS Reports:

- Portfolio vintage analysis (performance by booking month/quarter)
- Cohort delinquency evolution (how do Q1 2024 bookings age vs Q2?)
- Seasonal adjustment factors (Q1 = 1.35x default multiplier, discovered by YOU)
- Probability of Default (PD) modeling by segment

Senior Data Scientist integrates SAS outputs → Silver layer → Power BI feeds

GOOGLE BIGQUERY (BI Analytics Layer)

YOUR Role: Define aggregation tables, set refresh schedule, optimize for Power BI

Daily 10 AM IST sync:

- Snowflake Gold tables → BigQuery (incremental refresh)
- Pre-aggregated tables for dashboard performance:
    - Portfolio health (by product, region, bucket)
    - Collection metrics (by collector, team, outcome)
    - Delinquency trends (by month, product, FICO segment)

Refresh logic:

sql
```
-- Daily 10 AM IST (1:30 AM ET)
MERGE `bmo-analytics.credit_risk.portfolio_health` T
```

```sql
USING (
  SELECT
    DATE(snapshot_date) AS snapshot_date,
    product_type,
    SUM(outstanding_balance) AS total_balance,
    COUNT(DISTINCT account_id) AS account_count,
    AVG(days_past_due) AS avg_dpd,
    ... [BCR, delinquency %, etc.]
  FROM `snowflake-project.silver.account_delinquency_current`
  WHERE DATE(snapshot_date) = CURRENT_DATE()
  GROUP BY snapshot_date, product_type
) S
ON T.snapshot_date = S.snapshot_date AND T.product_type =
S.product_type
WHEN MATCHED THEN UPDATE SET
  T.total_balance = S.total_balance,
  T.account_count = S.account_count,
  ...
WHEN NOT MATCHED THEN INSERT *;
```

POWER BI (Visualization & User Interface)

YOUR Role: Design data model, create 20+ dashboards, set refresh schedules, manage 100+ users

Daily Refresh: 12 PM IST (1:30 AM ET)

- Import from BigQuery
- Refresh all 20+ production dashboards
- Validate calculations vs SQL
- Alert if BCR < 3.5 or Delinquency > 15%

DAX Measures (80+ across all dashboards):

- Application metrics: Total Apps, Approval Rate, Processing Days
- Portfolio metrics: Total Accounts, Current/Delinquent/CO splits, BCR, GLPA, BLPA
- Collection metrics: Recovery Rate, Cure Rate, Collector Performance (risk-adjusted)
- Executive KPIs: Portfolio Health Score, Monthly Revenue Impact

📅 DAILY/WEEKLY/MONTHLY WORKFLOWS

DAILY OPERATIONS (Monday–Friday, 30% of your time)

YOUR Daily Work Schedule (IST = India Time, ET = Eastern Time)

6:30 PM IST / 8:00 AM ET - Morning Dashboard Check

- Connect to Power BI Service
- Review yesterday's refresh status (99.5% SLA target)
- Check for data quality alerts (nulls, duplicates, variance)
- Confirm Alteryx workflow completed successfully

8:00 PM IST / 9:30 AM ET - Data Validation

- Validate BCR >= 3.5 (target range 3.5–4.5)
- Check Delinquency rate <= 15% (threshold alert if > 15%)
- Compare YTD portfolio balance vs yesterday (variance < 0.1%)
- Review application count vs daily expectation

10:00 PM IST / 11:30 AM ET - Stakeholder Support (Ad-hoc)

- Handle 3–5 daily requests from business users
- Portfolio drill-down (by product, region, FICO segment)
- Collection team: priority lists for today's contact targets
- Executive: KPI spot-checks, trend questions

12:00 AM IST / 1:30 PM ET - Collection Team Priority Lists

- Generate Top 100 delinquent accounts for ICU collection team
- Sorted by priority score (Balance × Bucket Weight × FICO Factor)
- Export to spreadsheet for distribution
- Alerts: Top movers (newly delinquent, recently cured)

WEEKLY DASHBOARDS & YOUR WORKFLOW

4-Week Rotating Cycle (Refresh every Monday 12 PM IST)

WEEK 1: ORIGINATION FOCUS

Dashboards:

- Origination Executive Dashboard (4 pages)

    - Page 1: KPI Overview (Total Apps, Approval Rate %, Booking Rate %, Avg Processing Days)
    - Page 2: Approval Rate by Product & Underwriter
    - Page 3: Declined Reasons Analysis
    - Page 4: Application Volume Trend

YOUR Weekly Work (Week 1):

- Monday 6:30 PM IST: Pull CAD data, validate application volumes
- Tuesday 8 PM IST: Calculate approval rates, booking rates by underwriter/product
- Wednesday 10 PM IST: Update dashboard, add weekly KPI cards & charts
- Thursday 12 AM IST: Stakeholder review with VP Underwriting
    - Present: Approval trends, declining apps, system vs manual approval split
    - Discuss: Policy adjustments, risk appetite changes
- Friday 2 AM IST: Email CRO with weekly origination summary
    - Top 3 KPIs: Application volume, Approval Rate, Avg Processing Time
    - Trend analysis vs last week & YoY

WEEK 2: PERFORMANCE FOCUS

Dashboards:

- Performance Portfolio Dashboard (5 pages)
    - Page 1: Portfolio Health (Total Accounts, Current %, Delinquent %, BCR Ratio)
    - Page 2: Delinquency Distribution (Current/X, X+1…X+5, Charge-Off)
    - Page 3: Bucket Transitions (Flow from X → X+1 → X+2)
    - Page 4: Product & Regional Comparison
    - Page 5: Cohort Analysis (by booking month/quarter)

YOUR Weekly Work (Week 2):

- Monday: Calculate delinquency buckets (X to X+5, CO) for all 30K accounts
- Tuesday: Compute BCR, GLPA, BLPA, cure rates, roll rates
- Wednesday: Update Performance Portfolio Dashboard
- Thursday: Bi-weekly sync with SVP Portfolio Management
    - Present: Portfolio health, delinquency trends, bucket movements, FICO migration
- Friday: Risk segmentation report (FICO bands, product comparison)

WEEK 3: COLLECTION FOCUS

Dashboards:

- Collection Scorecard Dashboard (4 pages) + Recovery Pipeline Dashboard (2 pages)
  - Collection Page 1: Recovery KPIs (Total Collected, Recovery Rate %, Cure Rate %)
  - Collection Page 2: Collector Performance (individual scorecards, risk-adjusted)
  - Collection Page 3: Agency Comparison (ICU vs Primary vs Tertiary)
  - Collection Page 4: Contact Activity (Call/Email/Letter breakdown, outcomes)
  - Recovery Page 1: Charged-Off Accounts (Total CO Balance, Recovery %)
  - Recovery Page 2: Settlement & Cost Analysis

YOUR Weekly Work (Week 3):

- Monday: Pull delinquent accounts (4,200), assign priority scores
- Tuesday: Generate collector scorecards (not just volume, but risk-adjusted metrics)
- Wednesday: Update collection dashboards
- Thursday: Agency performance review with VP Collections
  - Present: Team performance, recovery trends, compliance scores
  - Discuss: Staffing, training, process improvements
- Friday: Recovery pipeline update (CO accounts, settlement status)

WEEK 4: EXECUTIVE SUMMARY

Dashboards:

- Executive Summary Dashboard (2 pages) + Strategic Analysis Dashboard
  - Exec Page 1: Top 10 KPIs (all domains in single view)
  - Exec Page 2: Risk Flags & Alerts (BCR, Delinquency, Charge-offs, Seasonality)
  - Strategic: Trends, YoY comparison, projections

YOUR Weekly Work (Week 4):

- Monday: Consolidate monthly KPIs (applications, portfolio health, collections, recovery)
- Tuesday: Prepare trend analysis (what changed, why, forecasts)
- Wednesday: Build executive presentation (15–20 slides, board-ready)
- Thursday: Monthly review with CRO & CFO
  - Present: Monthly results, trends, strategic recommendations, risk alerts
  - Discuss: Capital allocation, policy changes, risk appetite

- Friday: Strategic deep-dive analysis
    - Example: Q1 seasonality (discovery: Q1 = 1.35x default multiplier)
    - Example: Cohort performance analysis
    - Example: Forecasting (approvals next month, delinquency trend)

MONTHLY DASHBOARDS (Refresh 1st of each month)

Board Report Dashboard - Executive summary for board meetings
Cohort Analysis Dashboard - Vintage performance by booking month/quarter
Product Performance Dashboard - Credit Card vs Personal Loan deep comparison

🎯 POWER BI BUILD GUIDE (Step-by-Step)

PHASE 1: DATA IMPORT & MODEL SETUP (Week 1–2)

Step 1.1: Import 12 Excel Files into Power BI

```text
Power BI Desktop → Home Tab → Get Data → Excel Workbook

Import Files (in order):
1. Branch_Master.xlsx       → Query: Branch (25 rows)
2. Underwriter_Master.xlsx  → Query: Underwriter (15 rows)
3. Collector_Master.xlsx    → Query: Collector (25 rows)
4. Customer_Master.xlsx     → Query: Customer (30,000 rows)
5. Bureau_Information.xlsx  → Query: Bureau (30,000 rows)
6. CAD_Master.xlsx          → Query: CAD (50,000 rows)
7. Account_Performance.xlsx → Query: Account_Performance (30,000 rows)
8. Payment_History.xlsx     → Query: Payment_History (100,000 rows)
9. Delinquent_Accounts.xlsx → Query: Delinquent (4,200 rows)
10. Collection_Activity.xlsx → Query: Collection (25,000 rows)
11. Chargeoff_Accounts.xlsx → Query: Chargeoff (750 rows)
12. Recovery_Activity.xlsx  → Query: Recovery (5,000 rows)

For each: Transform Data → Set correct data types:
- Dates: yyyy-MM-dd
- Amounts: Currency or Fixed Decimal
- IDs: Text
- Rates/Percentages: Decimal
```

Step 1.2: Create Date Table

```text
let
    StartDate = #date(2023, 8, 1),
    EndDate = #date(2025, 11, 30),
    Dates = List.Dates(StartDate, Duration.Days(EndDate - StartDate) +
1, #duration(1,0,0,0)),
    Table = Table.FromList(Dates, Splitter.SplitByNothing()),
    Renamed = Table.RenameColumns(Table, {{"Column1", "Date"}}),
    WithYear = Table.AddColumn(Renamed, "Year", each
Date.Year([Date])),
    WithMonth = Table.AddColumn(WithYear, "Month", each
Date.Month([Date])),
    WithMonthName = Table.AddColumn(WithMonth, "MonthName", each
Date.ToText([Date], "MMMM")),
    WithWeek = Table.AddColumn(WithMonthName, "WeekOfYear", each
Date.WeekOfYear([Date])),
    WithDayName = Table.AddColumn(WithWeek, "DayName", each
Date.DayOfWeekName([Date]))
in
    WithDayName

In Power BI: Mark this table as Date Table (Date column)
```

Step 1.3: Go to Model View → Create Relationships

From Table	From Column	To Table	To Column	Cardinality
CAD	customer_id	Customer	customer_id	Many-to-One
CAD	branch_id	Branch	branch_id	Many-to-One
CAD	underwriter_id	Underwriter	underwriter_id	Many-to-One
CAD	application_date	Date	Date	Many-to-One
Account_Performance	customer_id	Customer	customer_id	Many-to-One

Payment_History	account_id	Account_Performance	account_id	Many-to-One
Payment_History	payment_date	Date	Date	Many-to-One
Delinquent	account_id	Account_Performance	account_id	Many-to-One
Delinquent	collector_id	Collector	collector_id	Many-to-One
Collection	collector_id	Collector	collector_id	Many-to-One
Collection	account_id	Delinquent	account_id	Many-to-One
Collection	activity_date	Date	Date	Many-to-One
Chargeoff	account_id	Account_Performance	account_id	Many-to-One
Recovery	account_id	Chargeoff	account_id	Many-to-One
Recovery	activity_date	Date	Date	Many-to-One

PHASE 2: CREATE DAX MEASURES (80+ measures, 1–2 weeks)

Organize by category in separate measure tables:

Patient Origination Metrics (20 measures)

```text
Total Applications = COUNTROWS(CAD)

Approved Applications =
CALCULATE(COUNTROWS(CAD), CAD[application_status] = "Approved")

Approval Rate =
DIVIDE([Approved Applications], [Total Applications], 0)

Average Processing Days =
AVERAGE(CAD[processing_days])
```

```
System Approval % =
CALCULATE([Approved Applications], CAD[system_approval_flag] = TRUE) /
[Approved Applications]

Booking Rate =
CALCULATE(COUNTROWS(CAD), CAD[booking_status] = "Booked") / [Total
Applications]
```

[Copy all 20+ measures from comprehensive list in full BAU document]

Portfolio Performance Metrics (25 measures)

text

```
Total Accounts = DISTINCTCOUNT(Account_Performance[account_id])

Current Accounts =
CALCULATE([Total Accounts], Account_Performance[current_bucket] =
"Current/X")

Delinquent Accounts =
[Total Accounts] - [Current Accounts]

Delinquency Rate =
DIVIDE([Delinquent Accounts], [Total Accounts], 0)

Portfolio Balance =
SUM(Account_Performance[outstanding_balance])

GLPA =
CALCULATE(SUM(Account_Performance[outstanding_balance]),
Account_Performance[current_bucket] = "Current/X")
/ [Current Accounts]

BLPA =
CALCULATE(SUM(Account_Performance[outstanding_balance]),
Account_Performance[current_bucket] <> "Current/X")
/ [Delinquent Accounts]

BCR =
```

```
DIVIDE([GLPA], [BLPA], 0)
```

[+ 17 more measures: Cure Rate %, Roll Rate %, Charge-off Rate %, etc.]

Collection & Recovery Metrics (35 measures)

[All collection-specific measures + recovery measures]

PHASE 3: BUILD DASHBOARDS (2–3 weeks)

Dashboard 1: Origination Executive (4 pages, 50 visuals)

Page 1 Layout:

text

```
Header Row (KPI Cards):
  [Total Applications] [Approval Rate %] [Booking Rate %] [Avg
Processing Days]

Main Area (4 visuals, 2x2 grid):
  Top-Left: Line chart - Application Trend (monthly)
  Top-Right: Stacked Bar - Approval Rate by Product (CC vs PL)
  Bottom-Left: Pie - Application Status Distribution
  Bottom-Right: Table - Top 10 Underwriters by Volume

Right Sidebar (Slicers):
  - Product Type (dropdown)
  - Underwriter (multi-select)
  - Date Range (between)
```

[Repeat for all 4 pages of Origination + 5 pages Performance + 4 pages Collection + 2 pages Recovery + 2 pages Executive]

PHASE 4: SET REFRESH SCHEDULES

Power BI Service → Settings → Scheduled Refresh

- Daily: 12:00 PM IST (1:30 AM ET) - All 20+ dashboards
- Weekly Monday: 12:00 PM IST - Executive summary refresh

- Monthly 1st: 12:00 PM IST - Board report dashboard

📊 20 REALISTIC INTERVIEW Q&A (WITH YOUR SYNTHETIC PORTFOLIO)

Q1: Portfolio Scale & Accounts

Q: How many active accounts did you analyze in your Credit Card/Personal Loan portfolio?
A: About 30,000 active customer accounts—18,000 credit card, 12,000 personal loan. This is representative scale for segment-level retail analytics at a Tier 1 bank.

Q2: Annual Charge-Off Rate

Q: What was your portfolio's annual charge-off rate last year?
A: About 1.1% annually (~220–300 accounts). This closely mirrors industry averages for unsecured retail lending in Canada/US. Our SAS models tracked seasonal variance (Q1 ~1.35x higher).

Q3: 30+ DPD Delinquency

Q: What was your 30+ Days Past Due (DPD) delinquency rate?
A: Our 30+ DPD rate was approximately 14% across both portfolios, in line with market trends for credit-sensitive retail books.

Q4: Product-Level Delinquency

Q: By product, where was delinquency highest—credit cards or personal loans?
A: Delinquency was slightly higher in Personal Loans (~15–16%) vs Credit Cards (~13%), which is expected since unsecured loans carry higher inherent risk.

Q5: BCR (Balance Control Ratio)

Q: What was your portfolio's BCR—your critical risk metric?
A: Our BCR (Good Loan Per Account / Bad Loan Per Account) averaged 4.1, well within our target band of 3.5–4.5. We had alerts set to flag if BCR dropped below 3.5.

Q6: Application Volume & Approvals

Q: How many applications did you receive and what was your approval rate?
A: Of ~22,000 applications in a 12-month window, 87% were approved, 10% declined, 3% canceled. This is industry-consistent for unsecured lending.

Q7: System vs Manual Approvals

Q: What was your system-vs-underwriter approval split?
A: 80% of applications were approved automatically via risk engines, 20% routed to underwriter review due to exceptions or risk triggers.

Q8: Charge-Off Recovery Rate

Q: What's your average recovery rate on charged-off accounts?
A: We recovered about 19% of charged-off value within 12 months—strong vs. the typical 15–25% industry range. A mix of IRU Internal (higher rates) and third-party agencies.

Q9: Collection Contacts Per Account

Q: How many contacts per account did collections average before cure?
A: The collection team averaged 8.7 contact attempts per cured account (blend of calls, emails, letters). ICU team was more aggressive (9+), agencies lower (6–7).

Q10: Time-to-Cure

Q: What was your typical time to cure a delinquent account back to current?
A: Median time-to-cure was 17 days from 30+ DPD status. Credit cards cured slightly faster (~14 days) than personal loans (~20 days).

Q11: Daily Dashboard Monitoring

Q: How did you monitor portfolio health on a daily basis?
A: Every day at 12 PM IST (1:30 AM ET), dashboards auto-refreshed. I set alerts: if BCR < 3.5 or Delinquency jumped >15%, the team was immediately flagged with details.

Q12: FICO Risk Segmentation

Q: How did you map risk by FICO score?
A: FICO < 550 (high risk): 12%, 551–700 (medium): 67%, 701+ (low): 21%. We weighted approval strategies—higher buckets (700+) approved at 93%, lower at 68%.

Q13: Cure Rate & Improvement

Q: What was your quarter-over-quarter cure/improvement rate on delinquent buckets?
A: About 24% of delinquent accounts returned to current ("cured") each quarter after collection activity. Roll-forward (worsening) was ~18%, stable (no change) ~58%.

Q14: Payment Composition

Q: On collected payments, what was the principal-vs-interest-vs-late-fee split?
A: Approximately 65% principal, 25% interest, 10% late fees—this mix is typical for collections recovery flows in unsecured lending.

Q15: Dashboard Scope

Q: How many dashboards did you own and what was your refresh schedule?
A: I led 20+ production dashboards: 4–5 daily (health, collections priority, applications), 10 weekly (origination, performance, collection detail), 5 monthly (executives, board-ready). All refreshed on schedule; 99.5% uptime SLA.

Q16: Personal Contribution & Team Impact

Q: What was YOUR specific contribution vs. the rest of the team?
A: I architected the entire data model (star schema, 5 fact + 6 dimension tables), built 50+ DAX measures, deployed refresh automation, and mentored 5 analysts. Rahul was promoted to Mid-level in 6 months; Priya now leads a new collection optimization initiative.

Q17: Time Zone Alignment

Q: How did you align 9-hour time difference (India to Toronto) with live stakeholder expectations?
A: My core hours were 6:30 PM–2:30 AM IST (8 AM–4 PM Toronto ET). I often logged in 2–3 hours early/late as needed. Dashboards were ready ahead of their office hours; email summaries sent before morning meetings.

Q18: Biggest Discovery/Risk

Q: What was the biggest insight or risk you identified in your portfolio analysis?
A: Seasonality: Q1 approval rates dropped 9% and delinquency rose sharply (tax/refinancing effect). We adjusted models with a 1.35x Q1 risk multiplier—significant impact on capital planning.

Q19: Data Confidentiality

Q: How did you ensure synthetic data didn't disclose BMO's actual portfolio info?
A: All numbers are modeled from public/statistical sources—not BMO proprietary data. Portfolio size, delinquency rates, charge-off rates are representative industry ranges, not confidential internal figures. The focus was architecture and analytics logic, not specific volumes.

Q20: Building a New Dashboard Tomorrow

Q: If tasked to build a credit limit utilization dashboard in 24 hours, what's your process? A: (1) Define user story with CRO—KPIs, dimensions. (2) Write DAX: `Utilization % = SUM(current_balance) / SUM(credit_limit)`, by product/segment. (3) Prototype visuals (gauges, trends, drill-downs). (4) Get business feedback in same-day call. (5) Automate refresh, slicer integration by next morning. Delivery on time, quality assured.

🏦 BMO BANK OFFICE HOURS & YOUR SCHEDULE

BMO Bank HQ (Toronto, Canada)

- Office Hours: 9:00 AM – 5:00 PM Eastern Time (ET)
- Time Zone: EST (UTC-5) / EDT (UTC-4) depending on daylight saving

YOUR Work Schedule (India, IST = UTC+5:30)

Core Working Hours:

- 6:30 PM – 2:30 AM IST = 8:00 AM – 4:00 PM ET (Core overlap with BMO)
- Can extend 2–3 hours before/after as needed for stakeholder meetings

Typical Daily Timeline (IST):

- 6:30 PM IST (8:00 AM ET): Start, check overnight Alteryx/SQL runs, validate data
- 8:00 PM IST (9:30 AM ET): Stakeholder syncs begin (VP meetings, dashboard reviews)
- 10:00 PM IST (11:30 AM ET): Ad-hoc analysis, collection team support
- 12:00 AM IST (1:30 PM ET): Priority lists, CRO summaries
- 1:00 AM IST (2:30 PM ET): Final checks, wrap-up

Weekly Stakeholder Meetings (Convert to IST):

- Monday 11:30 AM ET = 9:00 PM IST – CRO weekly portfolio review
- Wednesday 2:00 PM ET = 11:30 PM IST – VP Collections sync
- Friday 10:00 AM ET = 7:30 PM IST – VP Underwriting origination review

Benefits of Your Schedule:

- Morning for BMO stakeholders (before daily meetings)
- Evening for YOU (natural work preference)

- Allows 2–3 hour flexibility for ad-hoc urgent issues

☑ YOUR ACHIEVEMENTS & BUSINESS IMPACT

Technical Accomplishments

✅ End-to-End Architecture – Unified 12 data marts into star schema ($12.7B visibility)
✅ $6M Revenue Opportunity – Identified through capital allocation analysis on risk-weighted assets
✅ 2% Data Accuracy – Improved from 28% error rate through standardization & validation
✅ 50+ DAX Measures – Comprehensive KPI library covering all 4 domains
✅ 20+ Production Dashboards – All at 99.5% uptime SLA
✅ 80% Faster Reporting – Automation of manual processes

Leadership & People

✅ Mentored 5 Analysts – Rahul promoted to Mid-level (6 months), Priya leads initiative
✅ 100+ Daily Users – Trained and supported across 3 departments
✅ Bi-Weekly Stakeholder Demos – CRO, SVP, VP-level credibility built

Operational Excellence

✅ Real-Time Dashboards – 15-min refresh vs 12-hour batch (previous)
✅ 25% Collection Efficiency – Automated priority scoring
✅ 30% Default Reduction Projected – Early warning system identified
✅ Seasonal Discovery – Q1 1.35x risk multiplier (input to risk models)

📝 CONCLUSION

This BAU document represents your complete 27-month credit risk analytics leadership at BMO Bank (via QUAUTO Technologies). Use it to:

1. Build the actual Power BI project – 12 Excel files are ready; follow PHASE 1–4 step-by-step
2. Demonstrate expertise in interviews – Reference 20 Q&A with realistic, defensible numbers

3. Mentor freshers – Hand them this guide + data files; they can rebuild the entire system
4. Build your portfolio – Take screenshots of dashboards, create a demo video

All data is synthetic but industry-representative. Your architecture, logic, and processes are based on real Capgemini work. You are ready.

Good luck with your Power BI project and interviews!

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